

# Routing-Signature Compression for Many-Testing-Safe Anonymous DHT Tuning

Replacing  $M$  by an effective multiplicity  $M_{\text{eff}}$

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## Abstract

Anonymous DHT tuning exposes a combinatorial explosion of knob settings. Naively treating each knob vector as a distinct hypothesis can make certified menus overly conservative.

We show that in many anonymous routing designs, a large policy pool collapses to a much smaller set of *distinct observable behaviors* on the support of contexts that actually occur. We formalize this via *routing signatures* and give a safe compression protocol that replaces the raw pool size  $M$  by an effective multiplicity  $M_{\text{eff}}$  inside family-wise error control, without weakening the guarantee. This is an interface note that isolates the compression step; Certified C shows how compressed menus plug into ongoing recertification under drift, with anytime-valid evidence and disagreement-gated change control.

**Scope.** We isolate and formalize *Routing-Signature Compression for Many-Testing-Safe Anonymous DHT Tuning* as a reusable contract surface in the anonymity / Anonymous-DHT series.

**Imports.** This paper owns the multiplicity reduction step for Certified A. For the baseline many-testing story and menu guarantee, see Certified A.[?] Signature-compression outputs are contract objects (Synthesis 0) and should be referenced by ABOM/OINL manifests (Anonymity C).[?, ?]

**Exports.** A routing-signature compression interface that reduces proof and log payload size by quotienting equivalent transcript views, while preserving the certified-menu guarantees.

**Artifact policy.** This archive is source-only; supporting artifacts are available on request. See [?].

## 1 Signatures

Policies differ in many internal details, but the adversary (and the privacy label) only depend on a chosen transcript projection  $X$ .

**Definition 1** (Routing signature). *Fix a context space  $\mathcal{C}$  (e.g., local load bucket, churn bucket, epoch index) and a transcript projection  $X$ . The signature of a policy  $\pi$  is the conditional law family  $\{\mathcal{L}_\pi(X \mid C = c) : c \in \mathcal{C}\}$ . Two policies are signature-equivalent if these conditional laws match on the support of observed contexts.*

**Remark 1** (Why signatures reduce multiplicity). *If two policies are signature-equivalent, then any statistical test or confidence bound that depends only on  $X$  (and  $C$ ) produces identical outcomes for the two policies. Counting both in a multiplicity correction is pure waste.*

## 2 Compression protocol

Given randomized logs (or a simulator), compute a finite approximation of signatures by hashing conditional histograms. A safe protocol is:

1. Choose a finite context partition  $\mathcal{C} = \bigsqcup_j \mathcal{C}_j$  and discretize  $X$ .
2. For each policy  $\pi_m$  and each bin  $\mathcal{C}_j$ , estimate the histogram of  $X$ .
3. Define a signature string by concatenating these histograms (with a fixed ordering and rounding rule) and hashing.
4. Let  $M_{\text{eff}}$  be the number of distinct signature hashes.
5. Run Certified A’s menu construction with  $M$  replaced by  $M_{\text{eff}}$  in the  $\alpha$ -splitting.

## 3 Why this is safe

**Lemma 1** (Multiplicity depends on distinct tests). *Any family-wise error control argument that uses a union bound (or a closed testing procedure) can replace a multiset of identical tests by a single representative without increasing error.*

The only subtlety is approximation error: histogram discretization can merge distinct signatures. This is handled by including a *verification margin*: only merge signatures if they are within a certified tolerance in TV (or if the discretization is exact by construction).

## 4 A tiny DHT-style example

Suppose contexts are “epoch parity” and  $X$  is “committee-contact count bucket.” A large knob grid (dozens of walk lengths, retry thresholds, and padding rates) can collapse to a handful of distinct contact-count distributions once discretized. In practice we have observed  $M_{\text{eff}} \ll M$  when knobs are redundant on the realized support of contexts.

## 5 Takeaways

- **Multiplicity should reflect distinct observable behaviors:** if two knob settings induce the same transcript law on realized contexts, they should not both count against the family-wise error budget.
- **Routing signatures are the quotient:** define signatures as conditional laws of the transcript projection  $X$  given context  $C$ , then compress by hashing discretized conditional histograms.
- **Safety requires a tolerance story:** only merge signatures when equality is exact by construction or when a certified TV tolerance is enforced; otherwise discretization can hide real differences.
- **Where it plugs in:** Certified A’s menu construction replaces  $M$  by  $M_{\text{eff}}$  after signature compression.

## References

- [1] Anonymous. Artifact and Disclosure Policy for the One-True-Archive (Source-Only Public Release). Series synthesis note (Synthesis 6), 2026. (This archive.)
- [2] Anonymous. Certified Menus for Anonymous DHT Lookups: Distribution-free parameter selection with simultaneous guarantees under many-testing. Certified series Paper A, 2026. (This archive.)
- [3] Anonymous. Contract Objects for Anonymous-DHT Privacy Budgets and Claims. Series synthesis note (Synthesis 0), 2026. (This archive.)
- [4] Anonymous. ABOM/OINL: Audit BOM and Outcome Identification Noise Label for Anonymous DHT Deployments. Anonymity series Paper C, 2026. (This archive.)